

JANNET AKANKSHA EKKA

Forward-Deployed AI Engineer • Multi-Agent Systems • AI-Native Delivery • Agent Evaluation and Security

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PROFESSIONAL SUMMARY

AI systems architect and operator who takes agentic systems from design to live production, architecture, evaluation gates, cloud deployment and 24/7 operations. Works **AI-native**: specifies the system, directs AI coding agents (**Claude Code**, CLI-first) to implement it, and stands as the **sole human-in-the-loop quality gate**, technical code audit, test review and manual merge before anything reaches the running system. Architect and operator of a **patent-pending multi-agent trading AI** (67K LOC, 2,815-test suite, 491 recorded findings, running continuously on Google Cloud) and of multi-agent assistants on Google ADK, MCP and AlloyDB. Brings **4+ years leading enterprise QA automation at Deloitte** to the problem most AI teams now face, **evaluating and monitoring agents**, and monitoring them: systems here refuse to act on data they cannot verify, and no substituted default passes without a warning. **Rank 1** PGP in AI/ML (UT Austin McCombs & Great Lakes). Deliberately lean, the whole platform is architected and operated from a single 16 GB laptop, and a **cuDF/RAPIDS rewrite** cut its validation pipeline **~6x on an NVIDIA T4**. Built while a full-time family carer.

TECHNICAL SKILLS

AI-Native Delivery	Claude Code (CLI & command-line environments), agent-directed development, spec-driven implementation, AI-generated hotfixes, automated Pull Request structuring, human-in-the-loop code audit & merge control, AI observability, terminal log diagnostics, defect triage, execution-error and systemic-drift isolation
GenAI & Agents	Google ADK, Model Context Protocol (MCP), Agent2Agent (A2A) SDK, Vertex AI, Gemini, Imagen, OpenServ SDK, multi-agent orchestration, prompt engineering, RAG & retrieval evaluation (recall@k, MRR, paired bootstrap), chunking, BM25, reciprocal-rank fusion, cross-encoder reranking, LangChain (via ADK), Hugging Face
ML & Quant	scikit-learn, TensorFlow, Keras, PyTorch, CatBoost, Optuna (TPE), contextual bandits, CPCV / Deflated Sharpe / PBO / FDR overfitting gates, conformal prediction, time-series forecasting, NLP & sentiment analysis, computer vision (ResNet, Fast R-CNN)
Cloud & Data	GCP. Cloud Run, Cloud Functions, Compute Engine, BigQuery (+ vector search & Conversational Analytics), AlloyDB & pgvector, Cloud SQL, Firestore, Pub/Sub, Cloud Scheduler, Cloud Build, Artifact Registry, Secret Manager, Vertex AI MLOps; AWS. S3, EC2, RDS, SageMaker, Textract, Comprehend; Cloudflare Workers
Engineering	Python, SQL, TypeScript, JavaScript; FastAPI, Flask, Streamlit, Django, React, Next.js; Docker, Postgres + pgvector, Git, Jenkins, GitHub Actions, systemd, pytest, Selenium; REST & WebSocket API integration
GPU & Acceleration	NVIDIA cuDF / RAPIDS (cudf.pandas), CUDA, Vertex AI GPU notebooks, T4 CPU-vs-GPU benchmarking, GPU-native pipeline rewrites
Analytics	pandas, NumPy, SciPy, Matplotlib, Seaborn, Tableau, Power BI, hypothesis testing, A/B testing, experiment design

FLAGSHIP PROJECT

Smart Money Trading (SMT). Explainable Multi-Agent Trading AI • Sole architect & operator <i>Patent pending. Indian provisional application No. 202631090789 (filed Jul 2026)</i> <i>Live: smt-weex-trading-bot.jannet-ekka.workers.dev • Hackathon build on Cloud Run: smt-world • Code: github.com/JannetEkka/smt-apac</i> - Architected a three-layer autonomous trading agent, quant base, learning loop and explanation layer, trading 8 crypto perpetual-futures pairs across 4 strategy horizons; 67,000 lines of Python, 2,815 tests, and a register of 491 measured findings. I specified it, AI coding agents wrote it, and nothing merged without my review. - Architected a 6-persona committee (order-flow, technical, whale/on-chain, sentiment, regime) aggregated by a learned Judge with quorum-renormalised conviction, so a dead data feed can never silently strand a decision threshold. - Built the learning loop end to end. Optuna/TPE optimiser, regime-aware contextual bandit for sizing, replay and reward shaping, and a weekly refit that ships only through a CPCV + Deflated-Sharpe + PBO + FDR overfitting gate; a per-pair P(up-in-4h) forecaster validated on non-overlapping windows reached AUC 0.723 on BTC, with gate-failing pairs auto-held in shadow. - Accelerated strategy validation with NVIDIA cuDF (cudf.pandas, zero code change) on a Vertex AI GPU notebook, a 2.5M-row CPCV pipeline dropped from ~24s to 3.9s (~6x) on a T4; the front end is a Three.js/WebGL world served from Cloudflare Workers. - Engineered a BigQuery data lake of 173K decisions and 447K klines with embeddings and vector search for pattern mining, plus a conversational analytics agent over a sanitised public view. - Operate it live on GCP. Compute Engine under systemd with watchdog and auto-restart, Cloud Run and Cloud Functions services, Secret Manager, a Streamlit dashboard and a Cloudflare-hosted user-facing UI; every trade ships a faithful ≤500-character "why" verified by a counterfactual faithfulness check, and a no-silent-fallback rule (any substituted default raises a warning and a liveness flag) that surfaced four multi-week data-integrity bugs. Stack: Python, Optuna, scikit-learn, CatBoost, BigQuery, Vertex AI & Gemini, Cloud Run, Cloud Functions, Secret Manager, Streamlit, Cloudflare Workers, Docker, pytest, GitHub Actions	Jan 2024 – Present
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SELECTED AI / ML PROJECTS

wiretrap.ai.studio · github.com/JannetEkka/wiretrap

- Built a security tool that checks which tools an agent actually called. Most scanners read the agent's reply and grade the text. This one reads the tool-call log.
- Every tool the agent declares gets cloned as a dead copy with the same name and JSON Schema. The copy records the call and returns fake data. Attacks run in separate Gemini conversations and a judge model reads the log.
- Caught a Google codelab agent saying "I don't have access or authorization to modify configuration files" after it had already run `ls -la` through its shell tool. Text scanners pass that. Wiretrap flags it and shows the arguments.
- A run with nothing classified reports **INCONCLUSIVE**, never **HARDENED**. Attacks against tools the agent doesn't have are dropped from the score. Approved fixes get re-run against the attacks that broke it, and the report says when one didn't hold.
- 8 attack categories, 9.4K lines of TypeScript, React and Express on Cloud Run. Vertex AI through Application Default Credentials with a four-model fallback. **No static API key anywhere in the app.**

Tech: TypeScript, React, Express, Vertex AI & Gemini, Cloud Run, Firestore, Secret Manager, Google AI Studio

SmartDesk. Multi-Agent Assistant with Evaluated RAG · Google Cloud Gen AI Academy capstone

2026

github.com/JannetEkka/smartdesk · Eval write-up: [evals/RESULTS.md](#)

- Built a root **ADK orchestrator** routing to specialised sub-agents (inbox, calendar, knowledge base) with a SequentialAgent composing the reply, on Gemini 2.5 Flash and Cloud Run. Gmail and Calendar are reached through **MCP servers written from scratch** against the protocol SDK over stdio, not off-the-shelf toolsets.
- Built a **RAG evaluation harness before touching retrieval**, 40 labelled questions over 120 notes, recall@k and MRR@k, significance by **paired bootstrap at 10,000 resamples**; production baseline recall@5 **0.963**, MRR@10 0.886. 54 tests across chunking, metrics, rerankers and integration.
- Compared six retrieval strategies, whole-note and chunked embedding, BM25, reciprocal-rank fusion, a local cross-encoder and a Gemini reranker. **The harness killed the change it was built to justify**: the cross-encoder won clearly on the development embedder (MRR@10 +0.094, p=0.04) then collapsed on the production embedder to +0.028 with a CI spanning zero. Shipping it would have added **190 MB of PyTorch and taken search from 29 ms to 2,382 ms** in an image deliberately cut from 1.8 GB to 340 MB, for a gain indistinguishable from noise.

Smart Money Tracker. Multi-Agent Blockchain Analytics · Hackathon winner

2024

- Orchestrated multi-agent whale tracking across **Ethereum, Polygon and BNB Chain** with the OpenServ SDK, using Alchemy webhooks for sub-second transaction monitoring on Cloud Run.
- Built a confidence-scored detector for accumulation, distribution and coordinated movements with automated exchange-wallet exclusion via Moralis labelling; backed by Cloud SQL + pgvector, Firestore and Gemini analysis.

VerseCanvas. AI Poetry-to-Art Generation Platform

2025

github.com/JannetEkka/versecanvas

- Engineered a **multi-stage generative pipeline**, semantic poem analysis with Gemini 2.0 → automated prompt engineering → image synthesis with **Imagen 3.0 on Vertex AI** → hybrid editing and text overlay (2,600 LOC across 6 modules).
- Extracts themes, mood, visual elements and narrative arc beyond literal text; 6 languages, 6 art styles, tunable mood intensity and multiple interpretations per poem, shipped as a Streamlit app.

Applied ML Portfolio · 13 end-to-end projects (UT Austin McCombs PGP)

2024 – 2025

jannetekka.github.io/DSPProjects · github.com/JannetEkka/DSPProjects

- **Automotive surveillance (capstone)**: ResNet50 make/model/year classification at 69.2% accuracy with Fast R-CNN detection over 16,185 images across 196 classes. **Customer churn**: 81.5% accuracy (Random Forest / Gradient Boosting). Also semiconductor yield prediction (591 sensor signals), SVHN digit and signal classification, financial-news sentiment with Mistral-7B embeddings, SQL business analytics, and K-Means + PCA-SVM vehicle segmentation.

PROFESSIONAL EXPERIENCE

Independent AI Infrastructure Operator & Developer — Smart Money Trading (SMT) Project

Jan 2024 – Present

- Deployed and maintain a proprietary, **high-availability multi-agent system** running 24/7 as a background system daemon on **GCP Compute Engine**.
- Manage end-to-end **AI observability**, defect triage through terminal log diagnostics to isolate execution errors and systemic drift.
- Engineered an **AI-native CI/CD workflow** using **Claude Code** via command-line environments to generate targeted hotfixes and automate Pull Request (PR) structures.
- Act as the **sole human-in-the-loop quality gate**, performing technical code audits and manual Git merges to protect system state integrity.
- **Won Best DeFi Application** at the OpenServ × Hack2skill hackathon (runner-up, Best Website Application) with the multi-agent blockchain analytics build that preceded SMT; later **#1 on the WEEX AI Wars I preliminary leaderboard** of 230+ teams.
- Also shipped VerseCanvas on Vertex AI and multi-agent assistants on Google **ADK / MCP / AlloyDB** to Cloud Run, all delivered **while a full-time family carer**, alongside the PGP in AI/ML at Rank 1 and the Google Cloud Gen AI Academy.

Deloitte Consulting LLP — Test Automation Engineer (Analyst)

Aug 2019 – Jan 2023 · Bangalore

- **Led a 6-member QA engineering team** delivering automated testing frameworks and enterprise data-processing solutions for Fortune 500 clients including AT&T Consumer Connect, Hewlett Packard Enterprise and The Dental Supply Company.
- Built Selenium and Katalon Studio automation components that **improved test execution efficiency by 83%**; Jenkins-based automated reporting cut manual analysis time by **75%**.
- Wrote Python analysis scripts and Tableau dashboards tracking quality metrics across **343 business components**, reducing test failures by 25%.
- Developed Python pipelines analysing **50,000+ SAP Hybris transactions** to isolate performance bottlenecks, and automated data validation for 10,000+ product configurations.

Tech: Selenium, Katalon Studio, Python, JavaScript, Jenkins, Salesforce, SAP Hybris, Tableau, JIRA, Git

- Engineered an interactive PDF processing system with coordinate-based text extraction, integrating **AWS Textract OCR** and Mistral models into an AI answer-evaluation workflow, and designed the ML data flow across S3, RDS and SageMaker.
- Built a bilingual (English/German) evaluation interface with real-time NLP feedback via Amazon Comprehend plus an analytics dashboard; scaled to **1,000+ concurrent submissions** and 100+ teacher sessions, cutting grading time 70% and lifting grading consistency 85%.
Tech: React, Next.js, TypeScript, Python, Django, AWS (EC2, S3, RDS, SageMaker, Textract, Comprehend)

Bachelor of Technology, Information Technology

2015 – 2019

Kalinga Institute of Industrial Technology (KIIT) University, Bhubaneswar. CGPA 7.17/10

ACHIEVEMENTS, RECOGNITION & CERTIFICATIONS

- **Provisional patent filed.** India, App. No. 202631090789 (Jul 2026), covering the SMT multi-agent decision architecture.
- **WEEX AI Wars I, #1 on the overall preliminary leaderboard** of 230+ teams and one of 37 finalists in the \$880K live AI trading competition; featured in [WEEX's official coverage](#) and two live AMAs.
- **Winner, Best DeFi Application.** OpenServ × Hack2skill hackathon; runner-up, Best Website Application.
- **Top 101 of 1,500+ teams.** Google Cloud Gen AI Academy APAC hackathon (Cohort 2), advanced to prototype refinement.
- **2 Spot Awards** and an **Applause Award** at Deloitte for project excellence (HPE Configure-Price-Quote).
- **Google Cloud Gen AI Academy.** Completion, May 2025 (ID 2025H2S04GENAI-A01917); **Gen AI Academy APAC 2026.** Cohort 1, Cohort 2 and Cohort 3 completions plus the Cohort 1 Hackathon.
- **Gen AI Exchange Hackathon 2025** (Google Cloud & Hack2skill), problem statement "Automating Test Case Generation with AI" · **Agentic AI Day · Asha AI Hackathon 2025** (JobsForHer Foundation).
- **CPSAT.** Certified Professional Selenium Automation Testing, 2022. All credentials verifiable on [Credly](#) and [Google Skills](#).